

Interview Assignment

The aim of this assignment is to solve a text classification problem. You should have two files - train_set.csv and test_set.csv, which are taken from a combination of the public basic company dataset (<https://data.gov.uk/dataset/4462e41a-7413-4359-97f5-420b6ca5f9c0/basic-company-data>) and synthetic data, which includes information on a variety of UK merchants. The files consist of three columns:

- desc - string description of a merchant
- MCC - merchant classification code, this tells us which category the merchant is from
- tag - the class that the desc belongs to

There 3 'tags'/classes that we have in this dataset

- 'evans' - This merchant is a clothing retailer (<https://www.evans.co.uk/>)
- 'b & m' - This merchant is a variety retailer (<https://www.bmstores.co.uk/>)
- 'other' - This class lumps anything other than 'b & m' and 'evans'

We like you to attempt the following questions:

1. Write Regex/rule-based approach to classify 'evans' and 'b & m' strings
2. Can you test the performance of this regex/rule-based approach on the training and test set? Give reasoning for the performance metric used
3. Can we do better with ML/NLP tools? Compute performance metric for both training and test set, and give reasoning for the performance metric used
4. Summarise your findings- which performs better and why?
5. Is there a technique we can use to discover other merchants in the 'other' class?

The assignment will need to be submitted in the form of a Jupyter notebook with outputs.